

Geometric Inequalities

Discussion of Geometric Inequalities usually about the following:

- Inequalities with geometric interpretation;
- Geometric Facts in the form of Inequalities;
- Geometric optimization.

Warm-up Exercise

If a, b, m and n are real numbers such that $a^2 + b^2 = 1$ and $m^2 + n^2 = 1$, prove that
 $|am + bn| \leq 1$. (Try to get as many solutions as you can)

A. Techniques, Classical Examples and Theorems

Example 1 (Isoperimetric Theorem for Triangle)

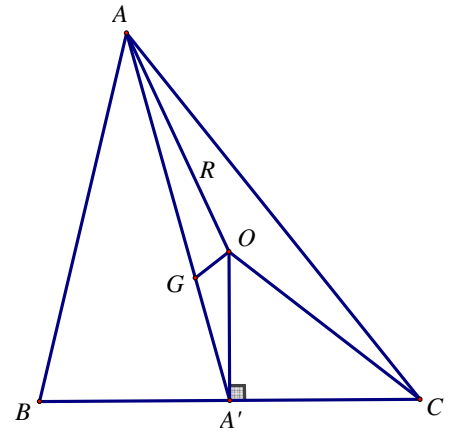
Among all triangles with a fixed perimeter, the equilateral triangle has the largest area.

Example 2 (Leibniz's inequality)

Prove the *Leibniz's theorem*: In $\triangle ABC$ with circumcentre O and centroid G , we have

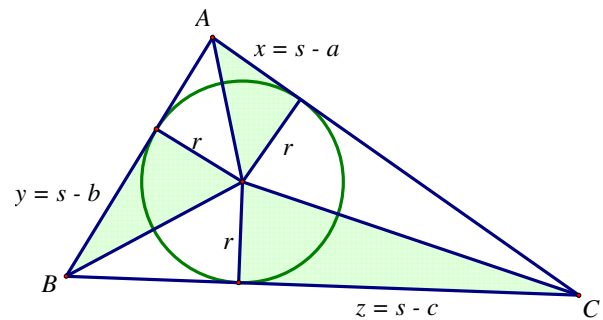
$$OG^2 = R^2 - \frac{1}{9}(a^2 + b^2 + c^2).$$

Hence deduce that $9R^2 \geq a^2 + b^2 + c^2$.



Ravi Substitution: $x = s - a, y = s - b, z = s - c$
i.e. $a = y + z, b = x + z, c = x + y$.

Ravi D Vakil
1970 -

**Example 3 (Padoa's Inequality)**

If a, b, c are sides of a triangle, prove that $abc \geq (a + b - c)(b + c - a)(c + a - b)$

Example 4 (Euler's Inequality)

For a triangle with circum-radius R and in-radius r , prove that $R \geq 2r$ and equality holds if and only if ABC is equilateral.

Example 5 (Ptolemy's Inequality)

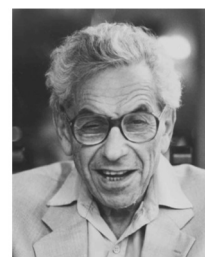
If $ABCD$ is a convex quadrilateral, then $AC \times BD \leq AB \times CD + BC \times DA$. The equality holds if and only if $ABCD$ is a cyclic quadrilateral.

Example 6 (Erdős-Mordell Inequality)

Given P is a point inside triangle ABC , $PA = x$, $PB = y$, and $PC = z$; u , v , w are the length of the perpendicular from P to sides a , b , and c respectively, then

$$x + y + z \geq 2(u + v + w).$$

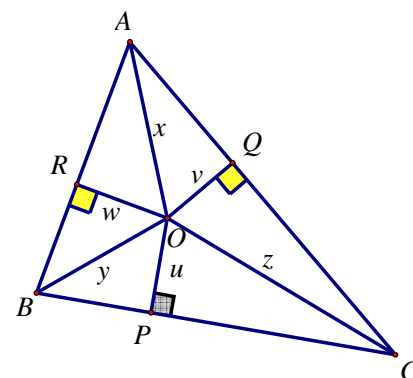
Equality holds if and only if ABC is equilateral and P is the centre of $\triangle ABC$.



Paul Erdős
(1913-1996)



Louis Joel Mordell
(1888-1972)



Example 7 (Fermat's Point)

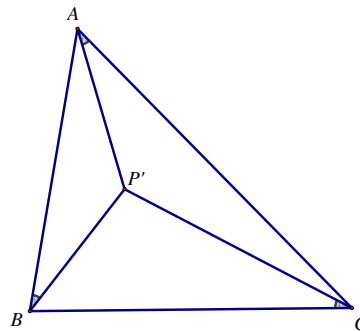
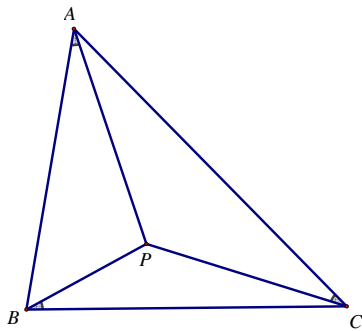
Given $\triangle ABC$, find a point P on the plane of ABC such that $PA + PB + PC$ is least.

The Brocard Points

For any triangle ABC , there exists a unique interior point P satisfying

$$\angle PAB = \angle PBC = \angle PCA.$$

The point P is called the first Brocard point of $\triangle ABC$.



Clearly, the isogonal conjugate P' of P , called the second Brocard point of $\triangle ABC$, is the unique interior point such that $\angle P'BA = \angle P'CB = \angle P'AC$.

The angle

$$\angle PAB = \angle PBC = \angle PCA = \angle P'BA = \angle P'CB = \angle P'AC = \delta$$

is called the Brocard angle of $\triangle ABC$.

The size of δ can be calculated from the formula

$$\cot \delta = \cot A + \cot B + \cot C = \frac{a^2 + b^2 + c^2}{4[ABC]}$$

Theorem

The greatest Brocard angle is 30° .

Proof: Follows immediately from the result of Problem 7 (IMO 1991 Question 5).

B. Problems**Problem 1**

If a, b, c are positive numbers, prove that $\sqrt{a^2 + ac + c^2} \leq \sqrt{a^2 - ab + b^2} + \sqrt{b^2 - bc + c^2}$.

Problem 2 [APMO 1996]

Let a, b, c be the lengths of the sides of a triangle, prove that

$$\sqrt{a+b-c} + \sqrt{b+c-a} + \sqrt{c+a-b} \leq \sqrt{a} + \sqrt{b} + \sqrt{c}.$$

Problem 3 [IMO 1983]

Let a, b and c be the lengths of the sides of a triangle. Prove that

$$a^2b(a-b) + b^2c(b-c) + c^2a(c-a) \geq 0.$$

Determine when equality occurs.



Problem 4 [IMO 1961] (Weitzenbock Inequalities)

Let a, b, c be the sides of a triangle, and T its area. Prove:

$$a^2 + b^2 + c^2 \geq 4\sqrt{3} T.$$

In what case does equality hold?

Problem 5 [IMO 1981]

P is a point inside a given triangle ABC . D, E, F are the feet of the perpendiculars from P to the lines BC, CA, AB respectively. Find all P for which

$$\frac{BC}{PD} + \frac{CA}{PE} + \frac{AB}{PF}$$

is least.

**Problem 6 [IMO 1991 Question 1]**

Given a triangle ABC , let I be the centre of its inscribed circle. The internal bisectors of the angles A, B, C meet the opposite sides in A', B', C' respectively. Prove that

$$\frac{1}{4} \leq \frac{AI \cdot BI \cdot CI}{AA' \cdot BB' \cdot CC'} \leq \frac{8}{27}.$$



Problem 7 [IMO 1991 Question 5]

Let ABC be a triangle and P an interior point of ABC . Show that at least one of the angles $\angle PAB$, $\angle PBC$, $\angle PCA$ is less than or equal to 30° .

**Problem 8 [IMO 1995]**

Let $ABCDEF$ be a convex hexagon with

$$AB = BC = CD,$$

$$DE = EF = FA,$$

and $\angle BCD = \angle EFA = 60^\circ$.

Let G and H be two points in the interior of the hexagon such that $\angle AGB = \angle DHE = 120^\circ$. Prove that

$$AG + GB + GH + DH + HE \geq CF.$$

